Twinstate

Judge & Audience Q&A — Presentation Prep Sheet

Your living 3D health twin: it turns the meds, painkillers, energy drinks and supplements a student stacks during exam season into a digital twin, screens the combination against HOLON's clinical drug-interaction knowledge, and explains in plain language what it's doing to their body.

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How to use this sheet. Answers are written to be spoken in 2–4 sentences. Lines marked SAY IT STRAIGHT are places where honesty wins points and overclaiming loses them — say the true version *before* a judge digs for it. Your single most important honest line: **real HOLON integration is wired in behind the service layer; when there's no key or network it runs on a deterministic clinical-rule engine so the demo never dead-ends.**

1 The 30-Second Basics

Q What is Twinstate, in one sentence?

A It's a living health twin for students: you log what you're taking, it builds a 3D body that visibly reacts, screens the combination against real clinical drug-interaction knowledge, and explains the risk in plain language with a clear path to care.

Q Who is it for?

A University students in exam season — the group most likely to stack caffeine, paracetamol, ibuprofen and "study aids" without realising the combined load. Secondarily, the campus health centre that sees the fallout.

Q What problem does it actually solve?

A Nobody checks the *combination*. Each thing feels harmless alone, but stacked together they strain the heart, liver, kidneys and gut. Twinstate is the missing safety check, made visual and personal.

Q Why is it called a "twin" and not just an app?

A Because it's organised exactly like the platform's digital twin: a body sliced into systems — brain, heart, liver, kidneys, stomach — each carrying the events (doses) and findings (flags) pinned to it. The body isn't decoration; it's the data model you can see.

2 Use of the Ontomorph Platform

Q Show me exactly where you use HOLON.

A Two calls on the real `@ontomorph/holon-client`: `concepts.search()` resolves every substance a student types to a real clinical concept, and `interactions.checkList()` screens the whole medication list against HOLON's interaction knowledge base. Severity, mechanism, clinical effect, management and evidence grade come back and drive our flags and citations.

Q What is real versus mocked right now? SAY IT STRAIGHT

A The HOLON integration is real and lives behind the service layer — it activates with a `holon_` key. When there's no key or the network is flaky, the app falls back to a deterministic clinical-rule engine so a demo never breaks. The DTP twin-core (grant-scoped patient data) is stubbed behind the same interface, ready to connect. I won't claim the offline rules are live HOLON output — they're the fallback, and HOLON slots into the identical interface.

Q Which parts of the platform did you use overall?

A Five: the digital-twin model, the 3D anatomy, HOLON's clinical knowledge, AI reasoning for the scores and explanations, and simulation for the two projected futures — one living body, end to end.

Q Why no patient grant / why didn't you use DTP patient data?

A Deliberate design. It's a self-service *consumer* tool, so it draws on HOLON's clinical *knowledge* (no grant required) rather than grant-scoped patient records. That means a student can use it the moment they open it. DTP is stubbed behind the same interface for when a clinic connects real records.

Q How much of HOLON's knowledge base can you reach?

A Concept resolution runs against HOLON's full concept space — millions of concepts — so an off-catalogue entry like "Augmentin" resolves to the real drug concept rather than being dropped.

Q If I gave you a live HOLON key right now, would it work?

A Yes — the client is wired in; setting `VITE_HOLON_API_KEY` switches the app from the fallback engine to live HOLON interactions and concept resolution without any code change.

3 Technical Architecture

Q What's the stack?

A React + Vite + Tailwind on the front end, with a clean service layer that abstracts three backends — HOLON, the AI reasoning layer, and the offline engine — behind one interface. State persists in the browser's local storage, so a student picks up where they left off.

Q What happens if the network or an API dies mid-demo?

A Nothing breaks. There's a graceful fallback engine and an error boundary, and the whole app runs fully on mocks with zero keys. A flaky venue network can't dead-end the demo — that was a deliberate execution goal.

Q Why build your own service layer instead of calling APIs directly?

A So each capability — HOLON, AI, offline — is swappable without touching the UI. It's what lets the same screens run live or offline, and it's where real DTP patient data will plug in later.

Q Is there a backend server?

A It's front-end-first with an optional lightweight proxy (`npm run proxy`) so an AI key never ships in the browser. Everything else runs client-side, which keeps the student's data on their device.

4 The 3D Anatomy

Q Is the 3D body real or is it a video / a picture?

A It's a live, dependency-free volumetric renderer — no heavy 3D library. Each organ is coloured and animated by the risk HOLON and the AI compute: a beating heart, a breathing chest, a scanning beam. You can drag to rotate and tap an organ to filter its flags.

Q Why build the 3D yourself instead of using a library?

A To keep it light, fast and fully controllable — the colours and animations are driven directly by our risk data, with no dependency to break at demo time. It's the emotional centrepiece, so it had to be reliable.

Q Tapping an organ does what?

A It focuses the twin on that system and filters the flag list to just that organ — so "why is my heart lit up?" is one tap from the specific interactions and substances behind it.

5 AI Reasoning

Q Where exactly is the AI, and what does it decide?

A The AI reasons over the twin's own data to calculate the per-organ scores, the 0–100 body-stress index, and the plain-language explanations, and it powers the grounded health chat. A deterministic engine provides an instant baseline and a fallback, and the UI even labels which source produced a number.

Q What stops the AI from hallucinating a drug interaction?

A The interactions themselves come from HOLON and the rule engine — *not* the language model. The AI's job is explanation, scoring and grounded conversation; it never invents the clinical facts. Answers also tag exactly what data they were grounded on.

Q Which model do you use?

A The reasoning and chat run through a configurable AI layer (an OpenAI-compatible key or a proxy URL). It's swappable — the app treats the model as a backend behind the service layer, and it degrades to the offline engine if no key is set.

Q If the AI is off, is the demo still meaningful?

A Yes. The offline engine still computes organ scores, the body-stress index and plain-language reasoning deterministically, so every screen is populated and truthful — you just lose the live-generated phrasing.

6 Clinical Validity & Safety

Q Are you giving medical advice? Isn't that dangerous?

A It's explicitly an educational tool, not a medical device — it doesn't diagnose or make prescription decisions. There's a consent gate, clear disclaimers, and emergency routing, and it always points to real care rather than replacing it.

Q What if a user is genuinely in danger during a chat?

A The chat has a hard safety rule. Red-flag phrases — chest pain, trouble breathing, overdose, self-harm and more — force an urgent "please seek care" response that routes to the campus health centre or emergency services, regardless of anything else it might say.

Q Where do your risk thresholds come from?

A Recognised public-health guidance: roughly 400 mg/day of caffeine and about 4 g/day of paracetamol as adult ceilings, NSAID-plus-dehydration renal stress, opioid-plus-sedative respiratory depression, SSRI-plus-tramadol serotonin risk. Each flag carries its source — NHS, FDA, EFSA.

Q Could this scare a user or push them to stop a real prescription?

A Severity is graded — calm, watch, caution, urgent — so the tone matches the actual risk, and the advice is "talk to whoever prescribed it," never "stop your medication." The framing is always plain language plus a path to care.

Q Are you storing health data? Is that a privacy/regulatory problem?

A Data stays in the student's own browser — there's no central health database to breach. That's a privacy-by-design choice that also keeps it out of heavier regulatory territory for a consumer educational tool.

Q Has a clinician validated this?

A Not formally — it's a hackathon build. The clinical logic is grounded in published FDA/NHS/EFSA guidance and, live, in HOLON's curated knowledge base, and clinician review is the first step before any real deployment. I'd rather be honest about that than overstate it.

7 What-If Simulation

Q What is the What-If coach doing?

A It projects two futures for the same body over the coming weeks — "keep this pattern" versus "rest + hydrate" — and shows the twin recolour and the body-stress index diverge, organ by organ, side by side. It turns a risk score into a trade-off you can watch.

Q Is the projection real modelling or just a curve?

A It's a transparent projection over the logged pattern, not a validated pharmacokinetic model — the point is to make the direction of travel visible and motivating. With live data and richer HOLON signals it can be tightened; today it's an honest, directional simulation.

Q Why two paths instead of a single prediction?

A Because the value is the contrast. A student doesn't need a precise number — they need to *see* that the same body cools down on one path and stays hot on the other. That comparison is what changes behaviour.

8 The Health Chat

Q How is the chat different from just asking ChatGPT?

A It's grounded in *this* student's twin — their actual logged intake, symptoms and flags — not generic web answers, and it tags exactly what it grounded each answer on. And it carries the same hard safety rule that routes anything urgent to real care.

Q What if someone asks about a drug you don't have in the catalogue?

A Unknown entries resolve live: HOLON first, then the AI, then a generic fallback — so an off-catalogue substance still gets identified and folded into the analysis rather than ignored.

9 Product, Market & Business

Q What's the business model?

A Free for students; the buyer is the university health service, which carries the cost and risk of student self-medication. B2B2C through campuses, with the consumer tool as the wedge.

Q Why students, why now?

A Exam-season polypharmacy is a real, common and under-checked harm, and students are digital-first and open to a phone-based tool. It's a sharp, underserved beachhead before expanding to anyone managing several substances at once.

Q Who are your competitors, and why are you different?

A Generic drug-interaction checkers exist, but they're a text box and a list. Twinstate is a living twin — a 3D body that reacts, two projected futures, and AI that computes the numbers from your own inputs. It's the difference between being told a risk and seeing it.

Q How would you get students to actually log honestly?

A Low-friction intake — rough doses are fine, resolves anything you type, under two minutes — plus the twin as an immediate, rewarding payoff. The visible reaction is the hook that makes logging worth it.

10 Roadmap

Q What would you build next if you won?

A Connect the live DTP patient twin for real records, bring in wearables for real-time vitals, widen coverage across more of HOLON's concepts, and add a clinician-facing view for the campus health centre.

Q How would you get this to production safely?

A Clinician review of the interaction logic first, then a formal validation pass against HOLON's knowledge base, a proper consent and data-governance layer, and a pilot with one campus health service before scaling.

11 The Skeptical Gotchas

Q Isn't this just a drug-interaction lookup with a 3D skin?

A The lookup is table stakes — the difference is the living twin, the two projected futures, and AI computing your personal numbers from your own inputs. It turns a static risk score into something you watch your own body do, which is what actually changes behaviour.

Q Did you really build all this during the hackathon?

A Yes — the git history shows the timeline. It works end to end today, with the safety framework, the fallbacks and the error boundary in place, which is exactly why it demos reliably.

Q What's the weakest part of the project? SAY IT STRAIGHT

A The simulation is directional rather than a validated pharmacokinetic model, and the clinical logic hasn't had formal clinician sign-off yet. Both are deliberate next steps, not hidden gaps — and neither undermines the core, which is the real HOLON integration and the living twin.

Q Why should this win?

A It hits every judging axis: innovation (a body that visibly reacts, with two futures), clinical value (a real, common harm with a genuine interaction check and a path to care), execution (works end to end with safety and graceful fallbacks), and platform use (real HOLON, 3D anatomy, AI reasoning, all behind one clean layer).

12 Live-Demo Contingencies

Q (If the network drops mid-demo)

A "This is running on the offline engine right now — which is exactly the fallback that keeps it working anywhere. With a HOLON key it's live clinical data through the same screens."

Q (If the AI response is slow)

A "While that generates — notice the numbers are already there from the deterministic engine; the AI is refining the phrasing on top of real computed scores."

Q (If asked to try an unusual substance live)

A Type it in intake — it resolves via HOLON, then AI, then a generic fallback, so it'll be identified and folded into the twin on the spot.